Ensemble-Based Spatial Snow Water Equivalent Estimates Using the SWE Evaluation Tool for the Rio Grande Basin

March 01, 2026

Project Team:
Max Gersh[1], Logan Stephenson[2], Ross Palomaki[2], and Karl Rittger[2]

Contributor:
Noah Molotch[2]

[1] New Mexico Office of the State Engineer (NM OSE) Hydrology Bureau
[2] Institute of Arctic and Alpine Research (INSTAAR), University of Colorado Boulder
Work funded by: NASA Western Water Actions Office (WWAO)
Contact: Max.Gersh@ose.nm.gov

Introduction

No spaceborne sensor can directly measure snow water equivalent (SWE). Therefore, the spatial distribution of SWE must be estimated from remotely sensed observations combined with modeling approaches. Existing gridded SWE products exhibit documented accuracy limitations across both mountainous and non-mountainous regions, resulting in a wide range of SWE estimates for any given location. To address this uncertainty, the New Mexico Office of the State Engineer (NM OSE), in collaboration with the Institute of Arctic and Alpine Research (INSTAAR), developed the SWE Evaluation Tool (SWEET). SWEET integrates multiple suitable SWE data products into an ensemble framework, with the objective of producing near-real-time spatial SWE estimates for the Rio Grande Basin that are, on average, as accurate as or more accurate than any individual product alone.

Figure 1 shows the near-real-time (March 01, 2026) ensemble SWE estimates for the Rio Grande Basin. Subsequent sections of this report present detailed ensemble-based SWE maps, along with summaries by subbasin and elevation band. In addition to the ensemble results highlighted in this report, figures and tables for each individual SWE product are also provided for download to support transparency and further analysis. These data sets and visualizations are available on the GitHub repository.

For ease of navigation, hyperlinks are provided throughout this report to help readers quickly access basin-specific sections and supporting data tables.

Rio Grande Basin SWE Overview

Rio Grande Headwaters

Upper Rio Grande

Rio Grande-Elephant Butte (Middle Rio Grande)

About SWEET

Rio Grande Basin SWE Overview

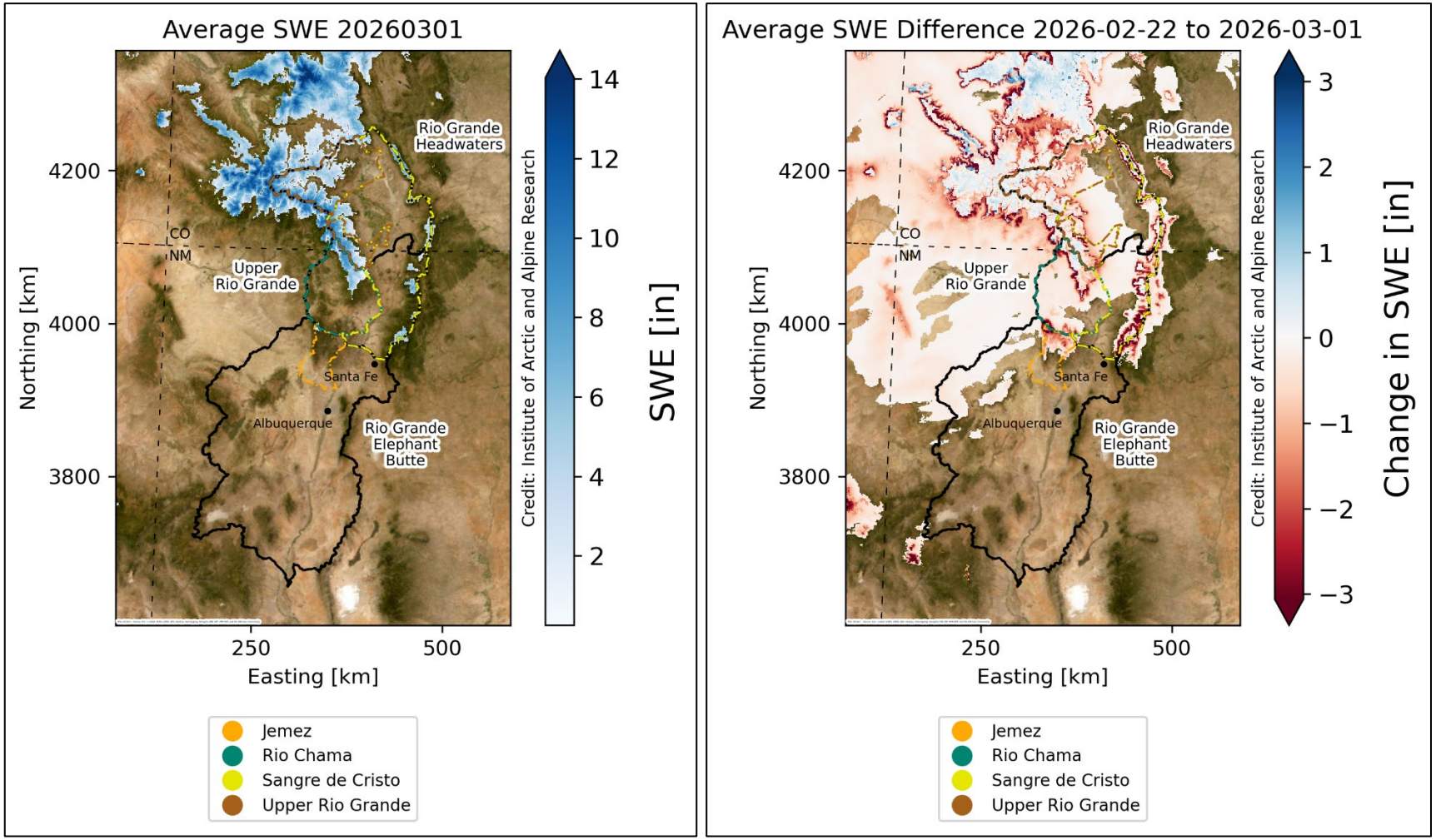

Figure 1. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Basin for March 01, 2026. *SWE amounts (in.) across the Rio Grande Basin (left); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Basin from February 22, 2026, to March 01, 2026 (right); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts.*

Table 1. Estimated ensemble SWE by HUC6 Basin for March 01, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 22, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each HUC8 basin (bold).*

HUC6 Basin	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-22 (af)	Median Vol. Change since 2026-02-22 (af)
Rio Grande Headwaters	Rio Grande Headwaters	337,744	336,621	315,295 (UA_SWE)	361,316 (CU_SWE)	-7,672	3,287
	Alamosa-Trinchera	106,655	112,126	94,996 (SNODAS)	112,842 (CU_SWE)	-24,472	-25,927
	San Luis	55,990	55,489	51,250 (CU_SWE)	61,230 (UA_SWE)	-7,863	-4,370
	Saguache	70,722	68,394	57,172 (SNODAS)	86,601 (UA_SWE)	-16,002	-13,218
	Conejos	145,354	136,801	136,547 (CU_SWE)	162,713 (SNODAS)	-7,904	-16,315
	Total	**716,464**	**712,053**	**706,991 (SNODAS)**	**730,349 (CU_SWE)**	**-63,913**	**-92,065**
Upper Rio Grande	Upper Rio Grande	85,214	90,479	46,042 (CU_SWE)	119,122 (SNODAS)	-37,955	-56,977
	Rio Chama	148,071	134,453	113,716 (UA_SWE)	196,043 (SNODAS)	-47,110	-47,825
	Total	**233,285**	**204,195**	**180,495 (CU_SWE)**	**315,165 (SNODAS)**	**-85,065**	**-125,538**
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	840	501	415 (UA_SWE)	1,603 (SNODAS)	-1,276	-1,497
	Jemez	5,738	6,850	3,045 (UA_SWE)	7,318 (CU_SWE)	-17,855	-16,943
	Rio Grande-Albuquerque	6	0	0 (SNODAS, UA_SWE)	19 (CU_SWE)	-89	-112
	Arroyo Chico	22	0	0 (SNODAS, UA_SWE)	65 (CU_SWE)	-3,515	-3,487
	North Plains	0	0	0	0	-2,661	-959
	Rio San Jose	118	0	0 (SNODAS, UA_SWE)	353 (CU_SWE)	-9,495	-12,084
	Plains of San Agustin	0	0	0	0	-1,437	-131
	Rio Salado	0	0	0	0	-290	0
	Jornada Del Muerto	0	0	0	0	0	0
	Elephant Butte Reservoir	18	0	0 (SNODAS, UA_SWE)	54 (CU_SWE)	-139	-221
	Rio Puerco	1,790	1,901	1,377 (UA_SWE)	2,092 (SNODAS)	-4,260	-5,077
	Total	**8,531**	**10,212**	**4,837 (UA_SWE)**	**10,546 (SNODAS)**	**-41,017**	**-48,177**

Table 2. Estimated ensemble SWE by specified subbasin groupings for March 01, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 22, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping (bold).*

OSE Grouping	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-22 (af)	Median Vol. Change since 2026-02-22 (af)
Upper Rio Grande	Rio Grande Headwaters	337,744	336,621	315,295 (UA_SWE)	361,316 (CU_SWE)	-7,672	3,287
	Saguache	70,722	68,394	57,172 (SNODAS)	86,601 (UA_SWE)	-16,002	-13,218
	Conejos	145,354	136,801	136,547 (CU_SWE)	162,713 (SNODAS)	-7,904	-16,315
	Total	**553,820**	**556,506**	**538,697 (UA_SWE)**	**566,257 (CU_SWE)**	**-31,578**	**-35,328**
Sangre de Cristo	San Luis	55,990	55,489	51,250 (CU_SWE)	61,230 (UA_SWE)	-7,863	-4,370
	Alamosa-Trinchera	106,655	112,126	94,996 (SNODAS)	112,842 (CU_SWE)	-24,472	-25,927
	Upper Rio Grande	85,214	90,479	46,042 (CU_SWE)	119,122 (SNODAS)	-37,955	-56,977
	Total	**247,859**	**263,834**	**210,135 (CU_SWE)**	**269,608 (SNODAS)**	**-70,290**	**-95,905**
Rio Chama	**Total**	**148,071**	**134,453**	**113,716 (UA_SWE)**	**196,043 (SNODAS)**	**-47,110**	**-47,825**
Jemez	**Total**	**5,738**	**6,850**	**3,045 (UA_SWE)**	**7,318 (CU_SWE)**	**-17,855**	**-16,943**

Upper Rio Grande Snowpack Update Application

The OSE/ISC Upper Rio Grande Snowpack Update Application (Application) provides water managers and users with detailed SNOTEL-based SWE data organized by the same subbasin groupings shown in Table 2. The Application is updated regularly from December through May and compiles key SWE information from SNOTEL sites within each subbasin. In addition to the Upper Rio Grande groupings (Upper Rio Grande, Sangre de Cristo, Rio Chama, and Jemez), the Application also includes the San Juan River Basin, with data summarized for the Upper San Juan, Animas, and Piedra subbasins.

The Application can be viewed here[1].

[1] Please note that access to the Application may be restricted to employees of the NM OSE/ISC at this time. We are currently working on public access.

Rio Grande Headwaters

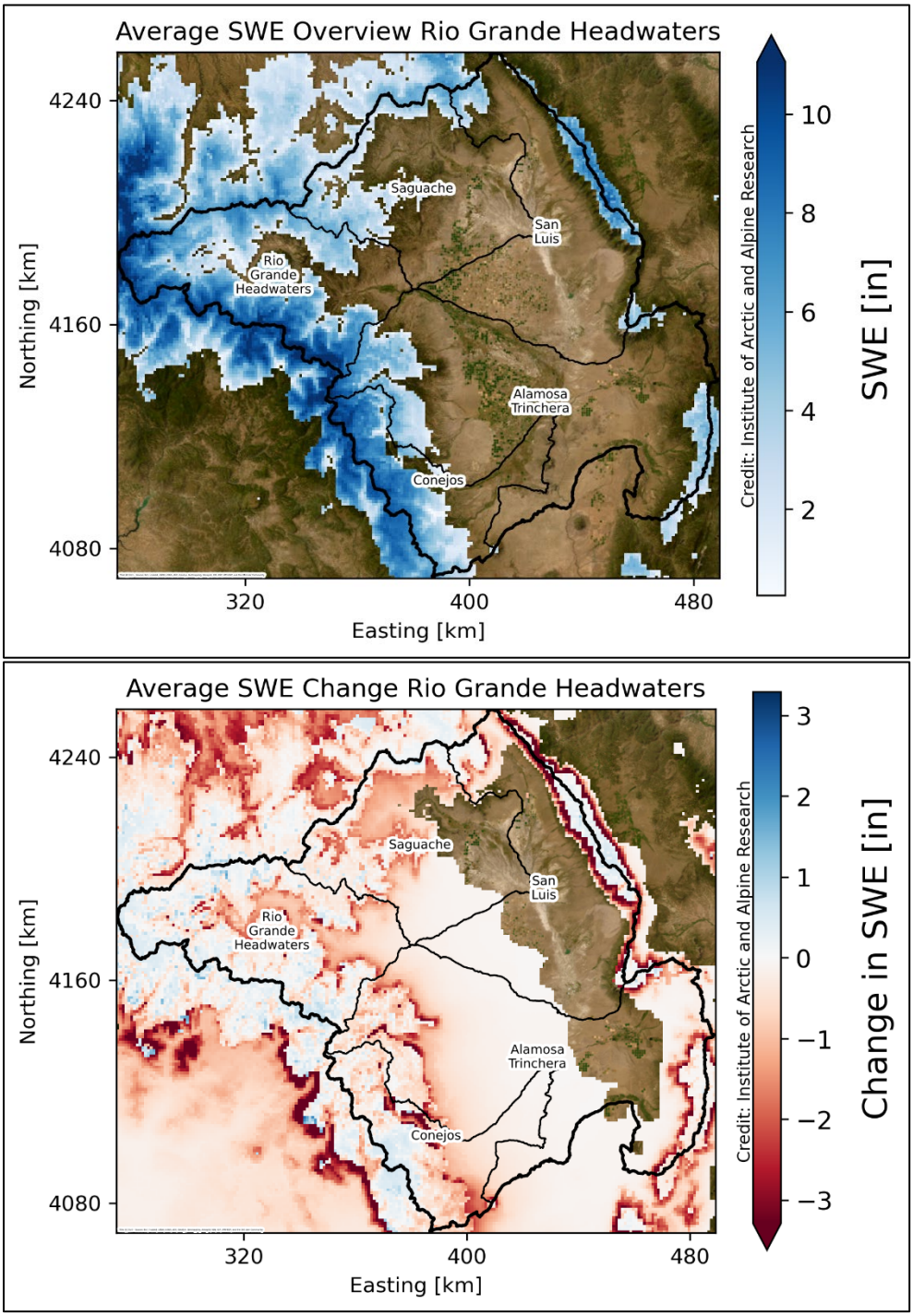

Figure 2. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Headwaters HUC6 Basin for March 01, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande Headwaters Basin, showing spatial patterns of SWE across subbasins including Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Headwaters from February 22, 2026, to March 01, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

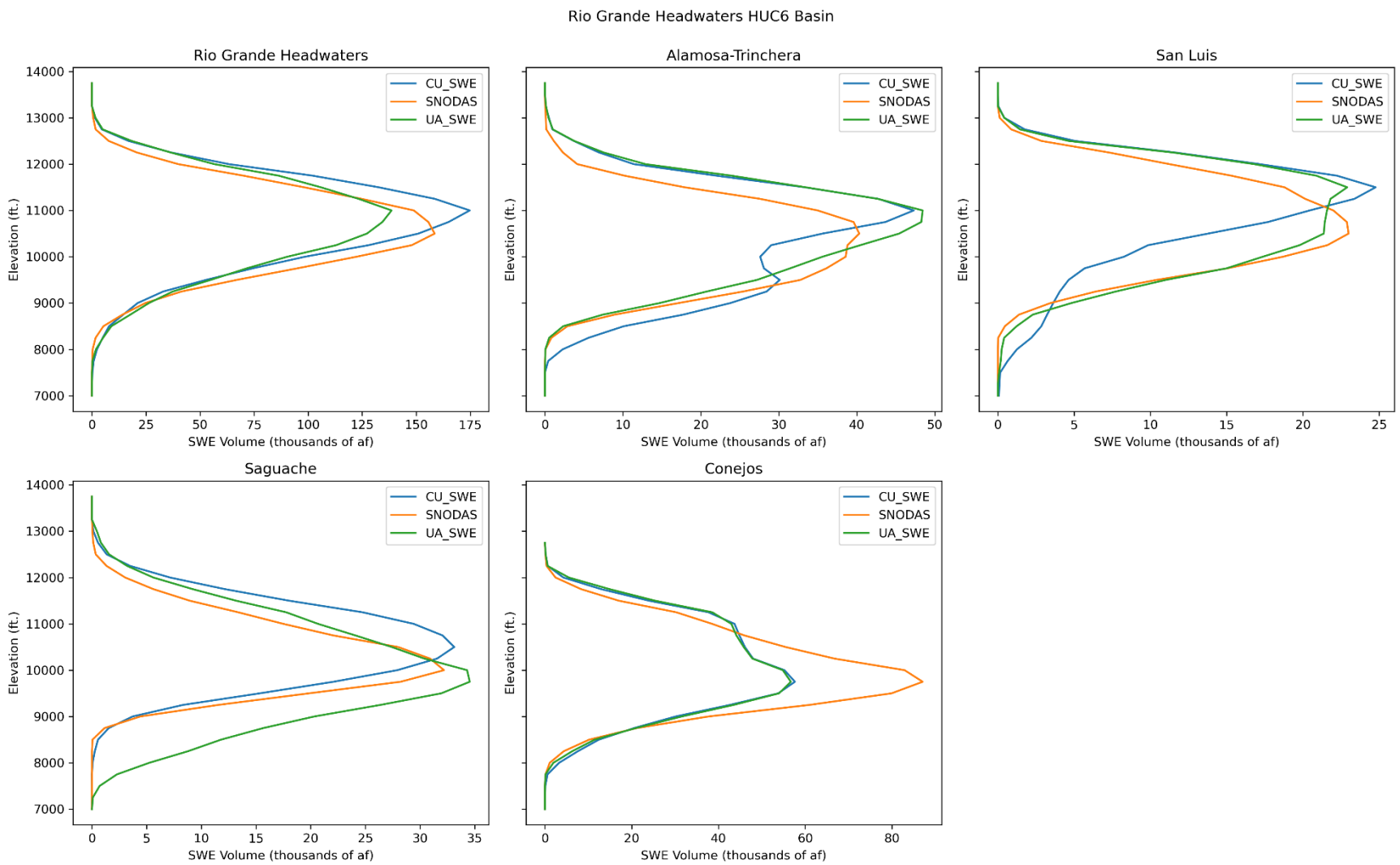

Figure 3. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for March 01, 2026. *Elevation-distributed SWE volume for the Rio Grande Headwaters HUC6 Basin and associated subbasins (Rio Grande Headwaters, Alamosa–Trinchera, San Luis, Saguache, and Conejos), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 3. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for March 01, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 22, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-22 (af)	Median Vol. Change since 2026-02-22 (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	0	0	0	0	-83	-115
		8000-9000'	1,474	1,822	301 (SNODAS)	2,298 (CU_SWE)	-5,283	-5,827
		9000-10000'	24,028	24,643	21,110 (CU_SWE)	26,332 (UA_SWE)	-7,559	-9,156
		10000-11000'	103,723	98,491	90,330 (UA_SWE)	122,349 (SNODAS)	1,986	3,444
		11000-12000'	154,085	148,891	138,599 (UA_SWE)	174,765 (CU_SWE)	2,569	-7,776
		12000-13000'	53,270	56,591	39,891 (SNODAS)	63,327 (CU_SWE)	715	463
		13000-14000'	1,164	1,325	546 (SNODAS)	1,622 (UA_SWE)	-16	92
	Alamosa-Trinchera	7000-8000'	0	0	0	0	-4,935	-4,620
		8000-9000'	794	66	51 (UA_SWE)	2,264 (CU_SWE)	-7,841	-7,044
		9000-10000'	18,539	17,139	14,707 (UA_SWE)	23,771 (CU_SWE)	-7,234	-7,193
		10000-11000'	33,918	35,583	27,606 (CU_SWE)	38,566 (SNODAS)	-2,492	-4,926
		11000-12000'	43,568	47,293	34,968 (SNODAS)	48,443 (UA_SWE)	-1,537	3,334
		12000-13000'	9,473	11,395	4,141 (SNODAS)	12,884 (UA_SWE)	-421	154
		13000-14000'	362	457	115 (SNODAS)	513 (CU_SWE)	-12	2
	San Luis	7000-8000'	25	0	0 (SNODAS, UA_SWE)	75 (CU_SWE)	-1,939	-1,400
		8000-9000'	497	247	0 (SNODAS)	1,244 (CU_SWE)	-2,620	-3,142
		9000-10000'	3,963	3,627	3,445 (SNODAS)	4,818 (UA_SWE)	-2,883	-3,367
		10000-11000'	14,772	17,363	8,267 (CU_SWE)	18,688 (SNODAS)	-1,023	-2,259
		11000-12000'	21,340	21,577	20,448 (CU_SWE)	21,994 (SNODAS)	361	938
		12000-13000'	15,074	16,808	11,246 (SNODAS)	17,168 (CU_SWE)	244	-56
		13000-14000'	318	417	116 (SNODAS)	422 (CU_SWE)	-5	3
	Saguache	7000-8000'	3	0	0 (SNODAS, UA_SWE)	8 (CU_SWE)	-1,114	-560
		8000-9000'	1,764	55	0 (SNODAS)	5,237 (UA_SWE)	-3,335	-2,025
		9000-10000'	9,484	4,413	3,712 (CU_SWE)	20,328 (UA_SWE)	-9,404	-12,801
		10000-11000'	31,471	32,191	27,920 (CU_SWE)	34,302 (UA_SWE)	-1,810	-4,616
		11000-12000'	22,530	20,672	17,497 (SNODAS)	29,420 (CU_SWE)	-198	198
		12000-13000'	5,267	5,615	3,039 (SNODAS)	7,146 (CU_SWE)	-136	112
		13000-14000'	204	131	33 (SNODAS)	448 (UA_SWE)	-4	3
	Conejos	7000-8000'	0	0	0	0	-719	-891

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-22 (af)	Median Vol. Change since 2026-02-22 (af)
Rio Grande Headwaters	Conejos	8000-9000'	2,104	1,946	1,103 (SNODAS)	3,264 (CU_SWE)	-3,821	-5,415
		9000-10000'	33,120	31,533	30,117 (CU_SWE)	37,711 (SNODAS)	-3,869	-7,488
		10000-11000'	64,319	55,211	54,835 (UA_SWE)	82,911 (SNODAS)	1,241	-2,346
		11000-12000'	41,741	42,975	38,543 (SNODAS)	43,706 (CU_SWE)	-662	113
		12000-13000'	4,069	4,250	2,445 (SNODAS)	5,513 (UA_SWE)	-74	327

Upper Rio Grande

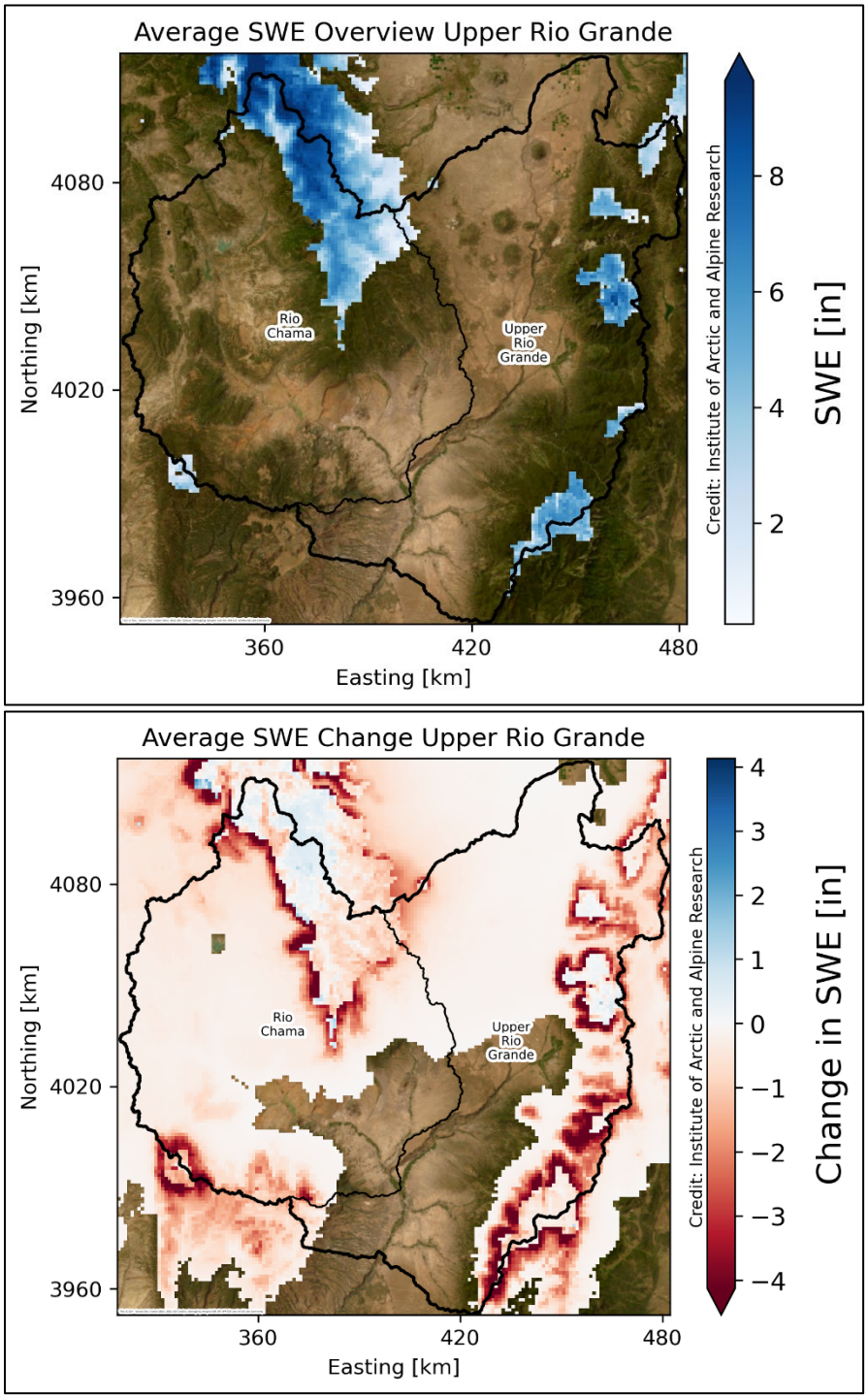

Figure 4. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Upper Rio Grande HUC6 Basin for March 01, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Upper Rio Grande Basin, showing spatial patterns of SWE across subbasins including Rio Chama and Upper Rio Grande (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Upper Rio Grande Basin from February 22, 2026, to March 01, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

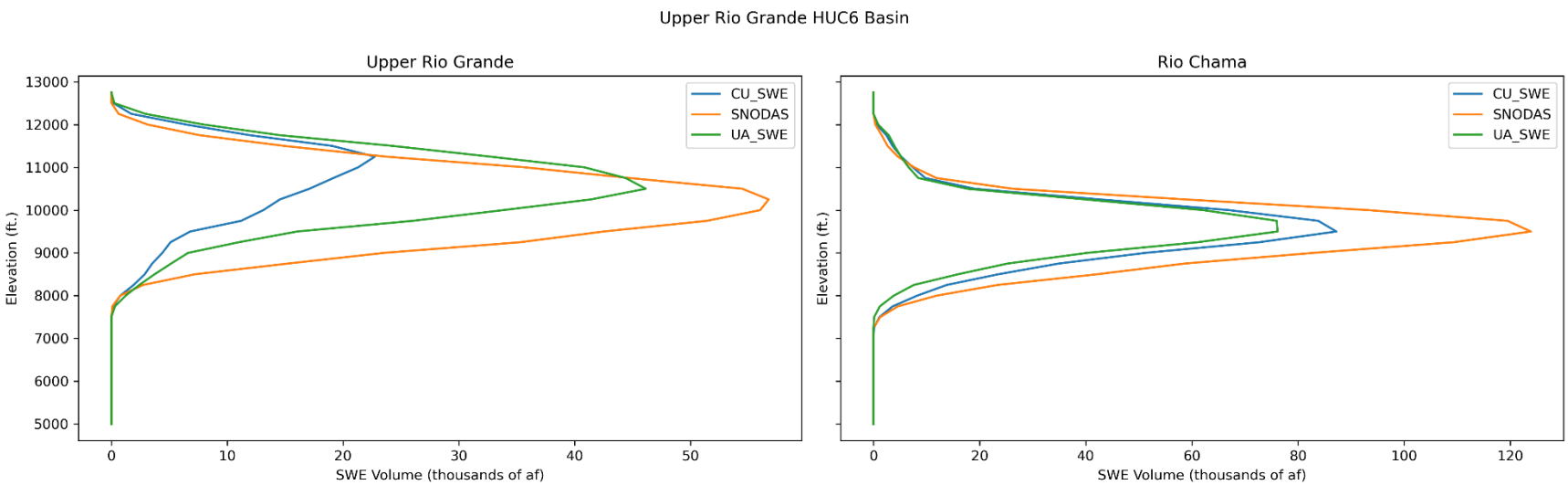

Figure 5. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Upper Rio Grande HUC6 Basin for March 01, 2026. *Elevation-distributed SWE volume for the Upper Rio Grande HUC6 Basin and associated subbasins (Rio Chama and Upper Rio Grande), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 4. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Upper Rio Grande HUC6 Basin for March 01, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 22, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-22 (af)	Median Vol. Change since 2026-02-22 (af)
Upper Rio Grande	Upper Rio Grande	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-115	-70
		7000-8000'	0	0	0	0	-4,243	-5,794
		8000-9000'	920	766	764 (SNODAS)	1,231 (UA_SWE)	-6,658	-8,487
		9000-10000'	11,512	6,621	4,399 (CU_SWE)	23,516 (SNODAS)	-15,636	-20,187
		10000-11000'	34,315	33,817	13,117 (CU_SWE)	56,011 (SNODAS)	-10,173	-16,949
		11000-12000'	32,614	35,701	21,303 (CU_SWE)	40,840 (UA_SWE)	-937	629
		12000-13000'	5,853	6,458	3,131 (SNODAS)	7,971 (UA_SWE)	-193	-248
	Rio Chama	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-820	-725
		7000-8000'	0	0	0	0	-12,439	-13,009
		8000-9000'	7,959	8,213	3,805 (UA_SWE)	11,859 (SNODAS)	-18,201	-13,532
		9000-10000'	58,111	51,212	40,249 (UA_SWE)	82,872 (SNODAS)	-14,938	-8,697
		10000-11000'	74,158	67,019	62,148 (UA_SWE)	93,307 (SNODAS)	-744	-1,450
		11000-12000'	7,232	7,414	6,623 (UA_SWE)	7,660 (SNODAS)	29	501
		12000-13000'	611	594	345 (SNODAS)	892 (UA_SWE)	3	55

Rio Grande-Elephant Butte (Middle Rio Grande)

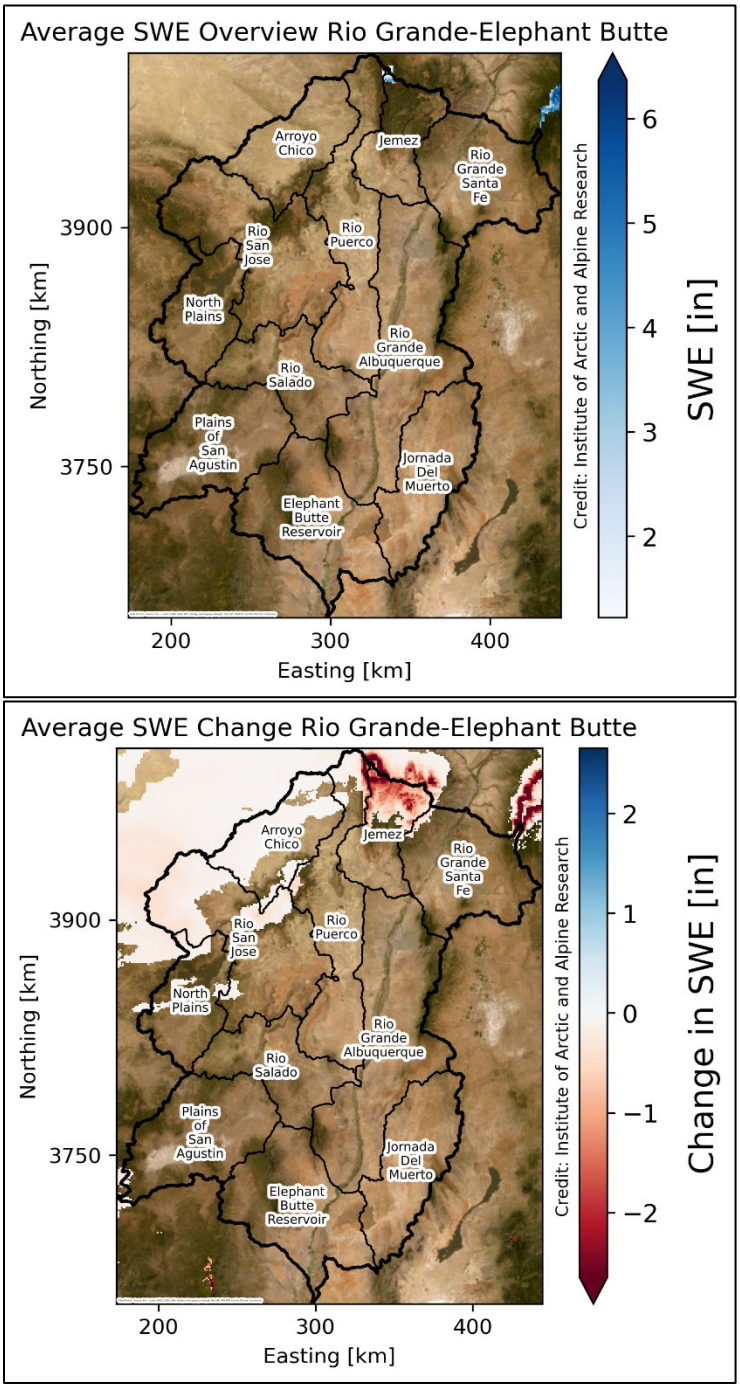

Figure 6. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande-Elephant Butte HUC6 Basin for March 01, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande-Elephant Butte Basin, showing spatial patterns of SWE across subbasins including Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande-Elephant Butte Basin from February 22, 2026, to March 01, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

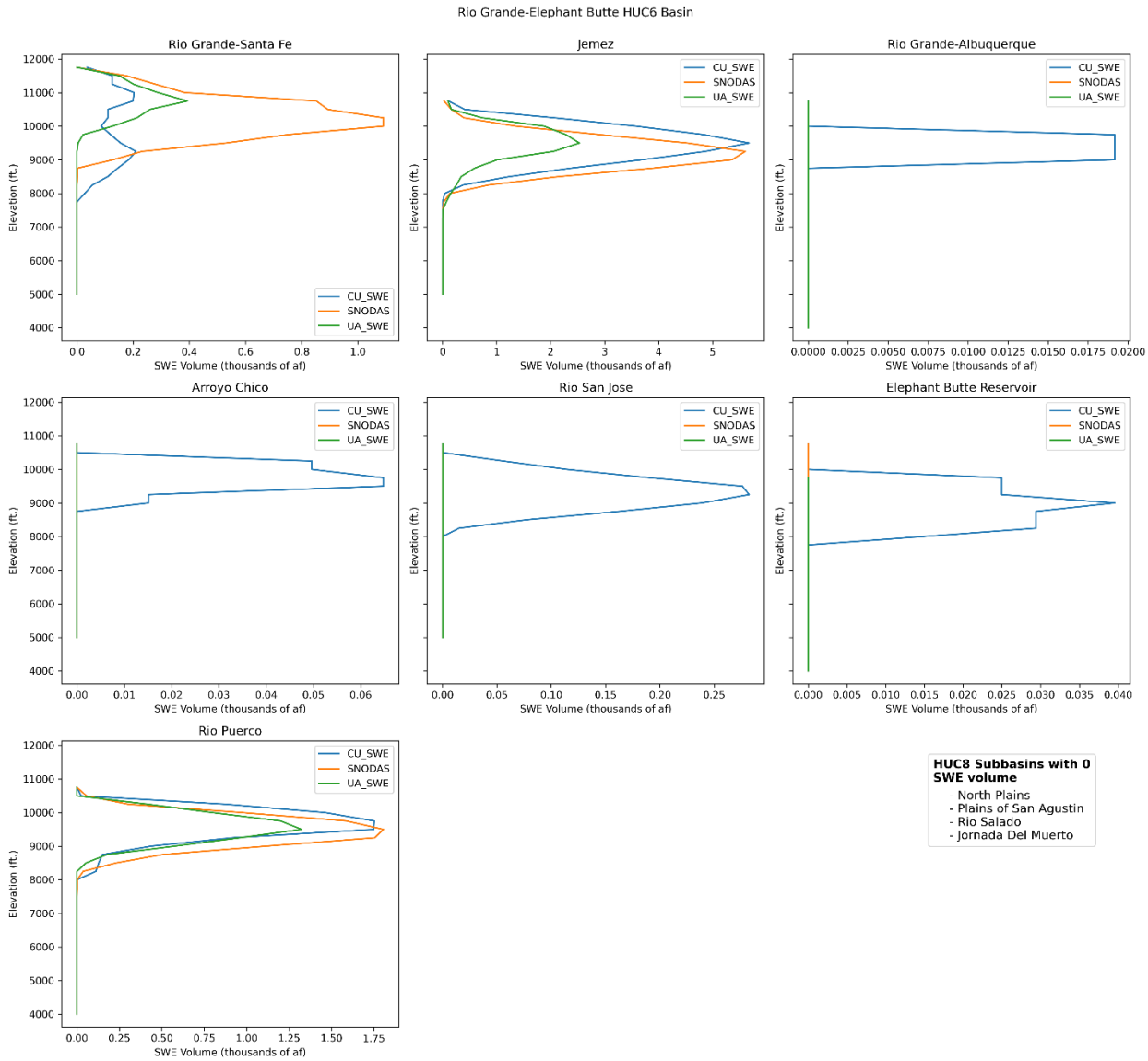

Figure 7. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for March 01, 2026. *Elevation-distributed SWE volume for the Rio Grande-Elephant Butte HUC6 Basin and associated subbasins (Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 5. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for March 01, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 22, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-22 (af)	Median Vol. Change since 2026-02-22 (af)
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-6	0
		7000-8000'	0	0	0	0	-113	-1
		8000-9000'	9	0	0 (SNODAS, UA_SWE)	27 (CU_SWE)	-452	-547
		9000-10000'	104	128	0 (UA_SWE)	184 (CU_SWE)	-550	-621
		10000-11000'	434	127	86 (CU_SWE)	1,090 (SNODAS)	-124	-275
		11000-12000'	292	288	203 (CU_SWE)	384 (SNODAS)	-32	-72
	Jemez	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-58	0
		7000-8000'	0	0	0	0	-512	-623
		8000-9000'	107	131	39 (CU_SWE)	151 (UA_SWE)	-9,617	-11,258
		9000-10000'	3,355	3,695	1,011 (UA_SWE)	5,360 (SNODAS)	-6,624	-4,934
		10000-11000'	2,276	1,883	1,360 (SNODAS)	3,584 (CU_SWE)	-1,045	-1,585
	Rio Grande-Albuquerque	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	-6	0
		8000-9000'	0	0	0	0	-56	-71
		9000-10000'	6	0	0 (SNODAS, UA_SWE)	19 (CU_SWE)	-26	-41
		10000-11000'	0	0	0	0	-2	0
	Arroyo Chico	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-1,485	-1,869
		7000-8000'	0	0	0	0	-1,053	-1,305
		8000-9000'	0	0	0	0	-872	-958
		9000-10000'	5	0	0 (SNODAS, UA_SWE)	15 (CU_SWE)	-91	-99
		10000-11000'	17	0	0 (SNODAS, UA_SWE)	50 (CU_SWE)	-14	-17
	North Plains	6000-7000'	0	0	0	0	-31	-46
		7000-8000'	0	0	0	0	-2,227	-432
		8000-9000'	0	0	0	0	-403	-481

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-22 (af)	Median Vol. Change since 2026-02-22 (af)
Rio Grande-Elephant Butte	Rio San Jose	5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-465	-575
		7000-8000'	0	0	0	0	-4,478	-6,194
		8000-9000'	0	0	0	0	-3,746	-3,736
		9000-10000'	80	0	0 (SNODAS, UA_SWE)	239 (CU_SWE)	-489	-513
		10000-11000'	38	0	0 (SNODAS, UA_SWE)	114 (CU_SWE)	-318	-188
	Plains of San Agustin	6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	-521	-19
		8000-9000'	0	0	0	0	-858	-110
		9000-10000'	0	0	0	0	-58	-2
	Rio Salado	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	-179	0
		8000-9000'	0	0	0	0	-108	0
		9000-10000'	0	0	0	0	-3	0
	Jornada Del Muerto	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
	Elephant Butte Reservoir	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	-0	0
		8000-9000'	5	0	0 (SNODAS, UA_SWE)	15 (CU_SWE)	-58	-39
		9000-10000'	13	0	0 (SNODAS, UA_SWE)	40 (CU_SWE)	-74	-98
		10000-11000'	0	0	0	0	-10	-10
	Rio Puerco	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	-284	-44
		7000-8000'	0	0	0	0	-887	-462

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-22 (af)	Median Vol. Change since 2026-02-22 (af)
Rio Grande-Elephant Butte	Rio Puerco	8000-9000'	1	0	0 (UA_SWE, CU_SWE)	2 (SNODAS)	-1,540	-2,018
		9000-10000'	711	577	438 (CU_SWE)	1,116 (SNODAS)	-1,162	-1,591
		10000-11000'	1,079	974	799 (UA_SWE)	1,463 (CU_SWE)	-387	-446

About SWEET

The following information provides a brief overview of SWEET, and the SWE products currently included in the ensemble framework. For more detailed descriptions, citations, and additional information, please refer to the GitHub repository.

SWE products currently used include:

1. **SNODAS SWE**
    a. The SNOw Data Assimilation System (SNODAS) is a physically based snow model developed by NOAA's National Operational Hydrologic Remote Sensing Center and operational since 2004. SNODAS produces daily 1-km gridded estimates of SWE and related snow variables by combining modeled weather inputs with satellite, airborne, and in situ snow observations.
2. **UA/SWANN SWE**
    a. The University of Arizona/Snow Water Artificial Neural Network Modeling System (UA/SWANN) is a research SWE product that uses snow models, artificial neural networks, and assimilated in situ measurements to generate daily gridded SWE and snow depth. The dataset provides 4-km and downscaled 800-m SWE across the conterminous United States, incorporating SNOTEL and COOP observations with PRISM precipitation and temperature fields.
3. **CU SWE**
    a. The University of Colorado (CU) SWE-Fusion product is an operational dataset produced by INSTAAR that provides near–real-time SWE estimates at 500-m resolution. The approach fuses SNOTEL observations with historical SWE reanalyses and terrain variables using a generalized linear model, and scales results using satellite-derived fractional snow-covered area from Snow Today.

The following notes describe key processing methods, assumptions, and visualization conventions used in generating SWE Evaluation Tool figures, tables, and elevation plots.

Figures:

- For ensemble figures, CU SWE and UA SWE are reprojected to the SNODAS resolution (1 sq km) and averages are only taken at pixels where at least 2 of the 3 products have SWE values greater than 0 inches.
- All overview and subbasin plots set pixels with values of 0 inches to NA so that the basemap is visible.
- For individual product figures, each product is displayed at its native resolution.
- Elevation bands of 250 feet are used in the elevation figures, and SWE volume is calculated for each product's native resolution in each band.
- Difference plots display pixels where snow was present in either the current or previous report.
- Figures are available in PNG format on the GitHub repository.

Tables:

- For all tables, mean, median, and volume are calculated at each product's native resolution - zeros are not removed from mean and median calculations.
- For ensemble tables, mean and median volume are calculated using the total volumes from the 3 products - not from an average aggregate raster.
- For min. and max., if 2 products have 0 volume, both products are listed in that cell. If all 3 products have 0 volume, no product is listed for min. or max. in that row.
- Change in SWE will only be calculated starting with the second report of the water year.
- Ensemble calculations: For HUC8 subbasins, the ensemble (mean and median) SWE and volume change (mean and median) are calculated individually for each subbasin. For the HUC6 "Total" row (tables 1 and 2), the ensemble is recalculated using the full HUC6 basin extent rather than summing HUC8. While the ensemble mean is additive and therefore consistent between HUC8 sums and the HUC6 total, the median reflects the middle value among the three products for the entire basin. As a result, median values will not sum to the HUC6 total, since they represent the difference between basin-wide medians rather than the sum of subbasin medians.
- Tables are available in CSV and PDF format on the GitHub repository.
- Elevation tables presented in the report (Tables 3, 4, and 5) are combined into one table (Table 3) on the GitHub repository.

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